

# ML DL

## PRACTICAL 2

Name : Parth Verma  
Class : D15C  
Roll no : 46  
Batch : B

### AIM :

To implement Multiple Regression , Ridge Regression and Lasso Regression on a real-life dataset.

### 1. Dataset Source

The dataset used for this experiment is the **Heart Disease Dataset**, obtained from Kaggle.

#### Source:

<https://www.kaggle.com/code/karim3ehab/automobile>

---

### 2. Dataset Description

The Heart Disease Dataset is a real-world medical dataset used to predict the presence of heart disease in patients based on various clinical attributes.

The dataset contains **1025 instances** and **14 attributes**, out of which **13 are input features** and **1 is the target variable**. The features include patient information such as age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, ECG results, maximum heart rate, and exercise-induced angina.

The **target variable** indicates whether a patient has heart disease:

- **Target = 1** → Presence of heart disease
- **Target = 0** → Absence of heart disease

The dataset consists of both **numerical and categorical features**, making it suitable for regression with regularization techniques.

---

### 3. Mathematical Formulation of the Algorithm

#### Ridge Regression (L2 Regularization)

For Ridge Regression:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

#### Lasso Regression (L1 Regularization)

For Lasso Regression:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

## 4. Algorithm Limitations

### Ridge Regression Limitations

- Does not perform feature selection
- All features remain in the model
- Less effective when many irrelevant features exist

### Lasso Regression Limitations

- Unstable when features are highly correlated
- May arbitrarily select one feature and discard others
- Performance depends heavily on the choice of ( $\lambda$ )

## 5. Methodology / Workflow

The experiment was conducted using the following steps:

1. Load the Heart Disease dataset
2. Separate features and target variable
3. Split the dataset into training and testing sets
4. Perform feature scaling using standardization
5. Train Ridge and Lasso regression models
6. Predict results on the test dataset
7. Evaluate model performance using error metrics
8. Compare coefficients and results

## 6. Performance Analysis

The performance of Ridge and Lasso regression models was evaluated using **Mean Squared Error (MSE)**.

- Ridge regression showed stable performance by reducing coefficient magnitudes and handling multicollinearity.
- Lasso regression produced sparse coefficients by eliminating less important features.
- Both models helped reduce overfitting compared to standard linear regression.

Lower MSE values indicate better model performance and improved generalization.

---

## 7. Hyperparameter Tuning

The regularization parameter (  $\lambda$  ) (alpha) was tuned to observe its effect on model performance.

- Different values of alpha were tested
- A higher alpha increased regularization strength
- Ridge regression benefited from moderate alpha values
- Lasso regression showed feature elimination at higher alpha values

Hyperparameter tuning helped in selecting optimal regularization strength, leading to improved prediction accuracy and reduced overfitting.

**OUTPUT :**

